آمار و احتمال مهندسی

Engineering Probability & Statistics

Lecture Notes ▪ جزوه درسی

Module 5

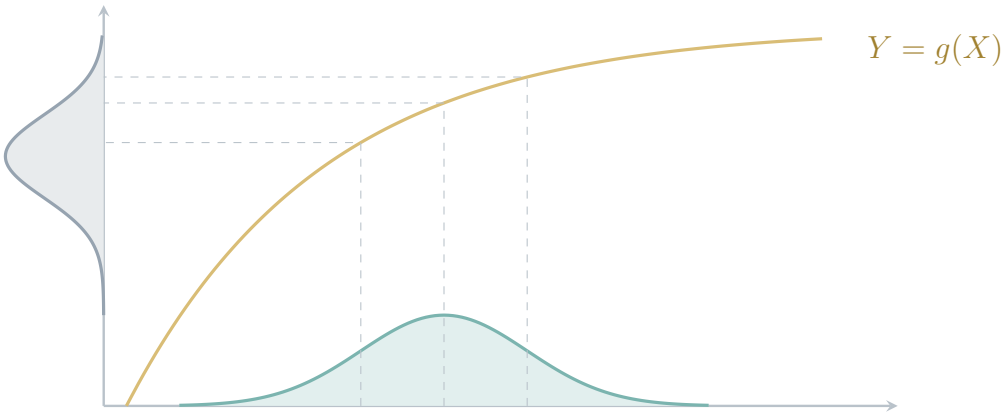
Important Random-Variable Functions

توابع مهم متغیر تصادفی

Instructor: Amin Azari

Compiled by: Aidin Shekari

استاد: امین آذری github.com/AminAzari



نیمسال پاییز ۱۴۰۶-۱۴۰۵

Fall Semester 2026-2027

In This Module

5.1 Introduction: Why Transform Random Variables? . . . 3
5.2 Linear Transformations: $Y = aX + b$ . . . 4
5.3 General Monotonic Transformations . . . 5
5.4 Nonlinear Transformations: Non-Monotonic Case . . . 6
5.5 The Jacobian Method . . . 7
5.6 Engineering Application: Rayleigh from Gaussian . . . 8
5.7 Practice Problems . . . 9

Learning Objectives

After completing this module, students will be able to:

1. Apply linear transformations to random variables
2. Derive PDFs of nonlinear transformations using the CDF and Jacobian methods
3. Apply transformations to engineering problems (power, SNR, envelope)
4. Verify analytical results with worked examples

5.1 Introduction: Why Transform Random Variables?

In engineering, we often know the statistics of one quantity but need the statistics of a **derived** quantity:

Known	Needed	Transformation
Voltage V	Power P	$P = \frac{V^2}{R}$
Linear SNR	SNR in dB	$Y = 10 \cdot \log_{10}(X)$
Gaussian I, Q	Envelope	$R = \sqrt{I^2 + Q^2}$
Current I	Power	$P = I^2 R$
Distance d	Path loss	$L = d^{\alpha}$

5.2 Linear Transformations: $Y = aX + b$

Result

If X has PDF $f_X(x)$, then $Y = aX + b$ has:

$$f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right)$$

Mean and Variance

- $E[Y] = aE[X] + b$
- $\mathrm{Var}(Y) = a^2\,\mathrm{Var}(X)$

Engineering Example: Amplifier

Input signal $X \sim N(0,\, \sigma^2)$ passes through amplifier with gain $G = 10$ and DC offset $V_0 = 1.5\,\mathrm{V}$.
Output: $Y = 10X + 1.5$

- $Y \sim N(10 \cdot 0 + 1.5,\, 10^2 \cdot \sigma^2) = N(1.5,\, 100\sigma^2)$
- Output noise power increased by $G^2 = 100$

Engineering Example: Temperature Sensor

Sensor output $V = 0.01 \cdot T + 0.5$ (V in volts, T in °C)
If $T \sim N(25,\, 4)$: $V \sim N(0.01 \times 25 + 0.5,\, 0.01^2 \times 4) = N(0.75,\, 0.0004)$

5.3 General Monotonic Transformations

CDF Method (Most General)

For $Y = g(X)$, find $f_Y(y)$ by:

1. Write $F_Y(y) = P(Y \leq y) = P(g(X) \leq y)$
2. Solve the inequality for X
3. Express in terms of F_X
4. Differentiate to get $f_Y(y)$

Formula for Monotonic g

If g is strictly increasing with inverse g^{-1}:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy}[g^{-1}(y)]$$

If g is strictly decreasing:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \left|\frac{d}{dy}[g^{-1}(y)]\right|$$

Combined (works for both):

$$f_Y(y) = \left.\frac{f_X(x)}{|g'(x)|}\right|_{x=g^{-1}(y)}$$

Engineering Example: SNR in dB

Linear SNR: $X \sim$ Exponential(1) (normalized Rayleigh fading power).
dB SNR: $Y = 10 \cdot \log_{10}(X)$

- $g(x) = 10 \cdot \log_{10}(x)$, $g'(x) = \dfrac{10}{x \cdot \ln 10}$
- $g^{-1}(y) = 10^{y/10}$

$$f_Y(y) = \frac{f_X(10^{y/10})}{10/(10^{y/10} \cdot \ln 10)} = \frac{\ln 10}{10} \cdot 10^{y/10} \cdot e^{-10^{y/10}}$$

This is NOT Gaussian — it has a longer left tail (deep fades).

5.4 Nonlinear Transformations: Non-Monotonic Case

When $Y = g(X)$ is Not One-to-One

If multiple x-values map to the same y, sum over all solutions:

$$f_Y(y) = \sum_i \frac{f_X(x_i)}{|g'(x_i)|}$$

where $x_1, x_2, \ldots$ are all roots of $g(x) = y$.

Engineering Example: Power from Voltage ($Y = X^2$)

Voltage $X \sim N(0, \sigma^2)$. Power: $Y = X^2$.
For $y > 0$, solutions are $x = +\sqrt{y}$ and $x = -\sqrt{y}$. $g'(x) = 2x$.

$$f_Y(y) = \frac{f_X(\sqrt{y})}{2\sqrt{y}} + \frac{f_X(-\sqrt{y})}{2\sqrt{y}} = \frac{1}{\sqrt{2\pi}\sigma} \cdot \frac{e^{-y/(2\sigma^2)}}{\sqrt{y}}$$

This is a **Chi-squared distribution with 1 degree of freedom** (scaled by σ^2).

Engineering Example: Full-Wave Rectifier ($Y = |X|$)

Input $X \sim N(0, \sigma^2)$. Output $Y = |X|$.
For $y > 0$: $x = +y$ and $x = -y$ are solutions. $|g'(x)| = 1$.

$$f_Y(y) = f_X(y) + f_X(-y) = \frac{2}{\sigma\sqrt{2\pi}} e^{-y^2/(2\sigma^2)}, \quad y \geq 0$$

This is the **folded normal** (half-normal) distribution.

5.5 The Jacobian Method

Procedure for $Y = g(X)$

1. Identify the transformation $y = g(x)$
2. Find the inverse: $x = g^{-1}(y)$
3. Compute the Jacobian: $J = |\frac{dx}{dy}| = |\frac{d[g^{-1}(y)]}{dy}|$
4. Apply: $f_Y(y) = f_X(g^{-1}(y)) \cdot |J|$

Step-by-Step Example: Exponential of Gaussian

$X \sim N(\mu, \sigma^2)$. $Y = e^x$ (log-normal transformation).

1. $g(x) = e^x$ (monotonically increasing)
2. $x = \ln(y)$, valid for $y > 0$
3. $J = |\dfrac{dx}{dy}| = \dfrac{1}{y}$
4. $f_Y(y) = f_X(\ln y) \cdot \dfrac{1}{y} = \dfrac{1}{y\sigma\sqrt{2\pi}} \cdot \exp\left(-\dfrac{(\ln y - \mu)^2}{2\sigma^2}\right)$

This is the **log-normal distribution** — models many engineering quantities (shadowing in wireless, stock prices, component lifetimes with wear).

5.6 Engineering Application: Rayleigh from Gaussian

Derivation of Rayleigh Distribution

In communications, the received signal has in-phase (I) and quadrature (Q) components:

- $I \sim N(0,\, \sigma^2)$, $Q \sim N(0,\, \sigma^2)$, independent

Envelope: $R = \sqrt{I^2 + Q^2}$

Using the transformation from $(I,\, Q) \to (R,\, \theta)$ with polar coordinates:

- $I = R \cdot \cos(\theta)$, $Q = R \cdot \sin(\theta)$
- Jacobian: $|\frac{\partial(I,Q)}{\partial(R,\theta)}| = R$

Joint PDF of $(R,\, \theta)$:

$$f_{R,\Theta}(r,\, \theta) = f_{I,Q}(r \cdot \cos\theta,\, r \cdot \sin\theta) \cdot r = \frac{r}{2\pi\sigma^2} \cdot e^{-r^2/(2\sigma^2)}$$

Marginalizing over $\theta \in [0,\, 2\pi)$:

$$f_R(r) = \frac{r}{\sigma^2} e^{-r^2/(2\sigma^2)}, \quad r \geq 0$$

This is the **Rayleigh distribution** — fundamental to wireless fading channel modeling.

5.7 Practice Problems

Problem 1

Signal $X \sim \text{Uniform}[0,\,1]$. Output $Y = -2 \cdot \ln(X)$. Find the PDF of Y and identify the distribution.

Solution CDF method: $F_Y(y) = P(-2\ln(X) \le y) = P(X \ge e^{-y/2}) = 1 - e^{-y/2}$ for $y \ge 0$. $f_Y(y) = \frac{1}{2}e^{-y/2} \to Y \sim \text{Exponential}(\beta = 2)$. (This is the inverse CDF method for generating exponentials!)

Problem 2

Noise voltage $X \sim N(0,\,\sigma^2)$ with $\sigma = 2\,\text{V}$. Power dissipated in $R = 50\,\Omega$: $P = \frac{X^2}{R}$.

(a) Find $E[P]$. (b) Find the PDF of P. (c) Find $P(P > 0.2\,\text{W})$.

Solution

(a) $E[P] = \dfrac{E[X^2]}{R} = \dfrac{\sigma^2}{R} = \dfrac{4}{50} = 0.08\,\text{W}$

(b) $P = \frac{X^2}{50}$. Let $W = X^2 \sim \sigma^2 \cdot \chi^2(1)$. Then $P = \frac{W}{50}$. $f_P(p) = 50 \cdot f_W(50p) = \frac{50}{2\sigma^2} \cdot \frac{50p}{\sigma^2}^{-1/2} \cdot e^{-50p/(2\sigma^2)}$ for $p > 0$.

(c) $P(P > 0.2) = P(\chi_1^2 > 2.5) = 1 - F_{\chi_1^2}(2.5) \approx 0.114$

Problem 3

$X \sim \text{Exponential}(\lambda = 1)$. $Y = \sqrt{X}$. Find $f_Y(y)$.

Solution $g(x) = \sqrt{x} \to x = y^2$, $\dfrac{dx}{dy} = 2y$.

$f_Y(y) = f_X(y^2) \cdot |2y| = e^{-y^2} \cdot 2y$ for $y \ge 0$. This is a Rayleigh distribution with $\sigma^2 = \frac{1}{2}$.

Problem 4

Distance $D \sim \text{Uniform}[1,\,10]$ km. Path loss: $L = 20 \cdot \log_{10}(D)$ dB. Find $E[L]$ and the PDF of L.

Solution $E[L] = E[20 \cdot \log_{10}(D)] = \displaystyle\int_1^{10} 20 \cdot \log_{10}(d) \cdot \frac{1}{9}\,dd = \frac{20}{9} \cdot \int_1^{10} \log_{10}(d)\,dd$

$= \frac{20}{9} \cdot \left[d \cdot \log_{10}(d) - \frac{d}{\ln(10)}\right]_1^{10} = \frac{20}{9} \cdot \left[10 - \frac{10}{\ln 10} + \frac{1}{\ln 10}\right] = \frac{20}{9} \cdot \left(10 - \frac{9}{\ln 10}\right) \approx 13.55\,\text{dB}$

PDF: L ranges from 0 to 20 dB. $g^{-1}(l) = 10^{l/20}$, $|\frac{dg^{-1}}{dl}| = \frac{\ln 10}{20} \cdot 10^{l/20}$.

$f_L(l) = \frac{1}{9} \cdot \frac{\ln 10}{20} \cdot 10^{l/20}$ for $0 \le l \le 20$.

Next Module: Module 6 — Two Random Variables